Concept Questions - Answers

Practice Part C - Questions 1-3

1. Unsorted versus sorted search
An unsorted array may require checking every element: worst-case O(n). A sorted array supports binary search, which discards half the remaining range per step and takes O(log n). This comparison assumes the array is already sorted; sorting first adds cost.

2. Why 2n is O(n)
Big-O ignores constant multipliers and describes asymptotic growth. The factor 2 changes only the constant amount of work, so 2n is O(n).

3. Linear versus non-linear traversal
Arrays arrange elements in a sequence, so traversal generally moves index by index. Trees branch through parent-child links and use strategies such as depth-first or breadth-first search to visit their branches.